Taghleef Industries

PACKAGING SOLUTIONS WORLDWIDE

Ti for shrink labels

AGENDA

Taghleef Industries

- **Shrink Market Overview**
- ***Ti* Shrink Film Product Overview**
- **Introduction to U002**

Ti : A wide product range for labels

Taghleef Industries

In-Mould

Self-Adhesive

WAL Reel-fed

WAL Cut & Stack

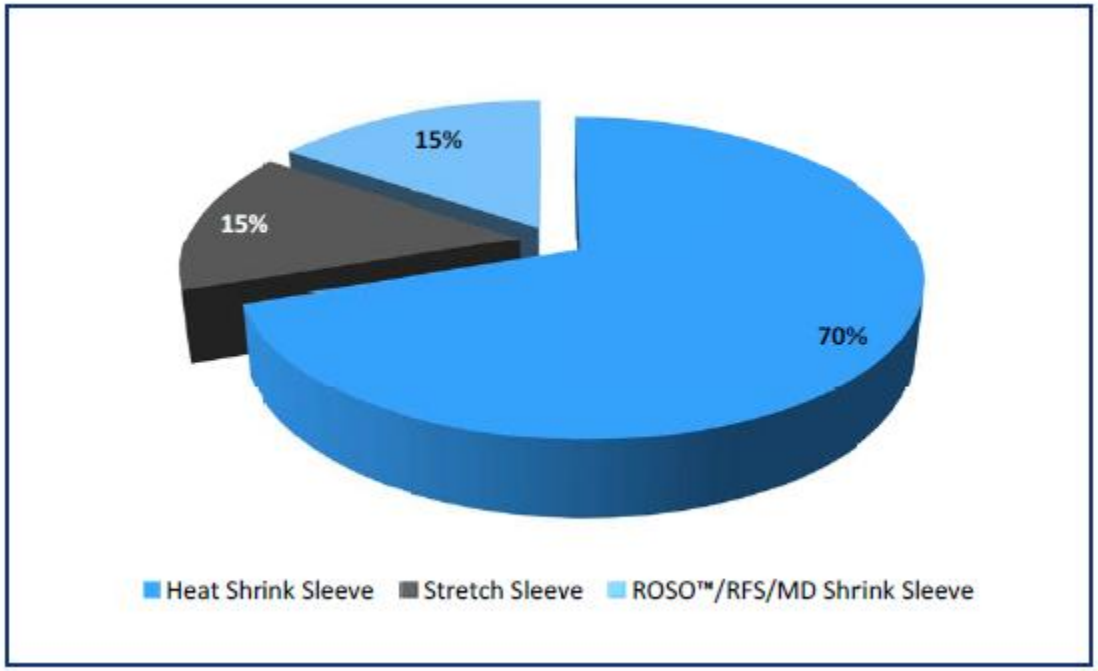
WAL Shrink - VISION®

NEW to EU

Market Overview
Shrink Label films

Taghleef Industries

North America



Source: AWA

- WAL Shrink = 15% share of contoured container labeling
- Two (2) Suppliers of WAL Shrink:
  - Ti = 85% share
  - ExMo = 15% share
- Key Market Applications:
  - Aerosol Cans
  - Coffee Cans
  - Powdered Drink Mix
  - PET Beverage

Our products and applications
Shrink Label films

Taghleef Industries

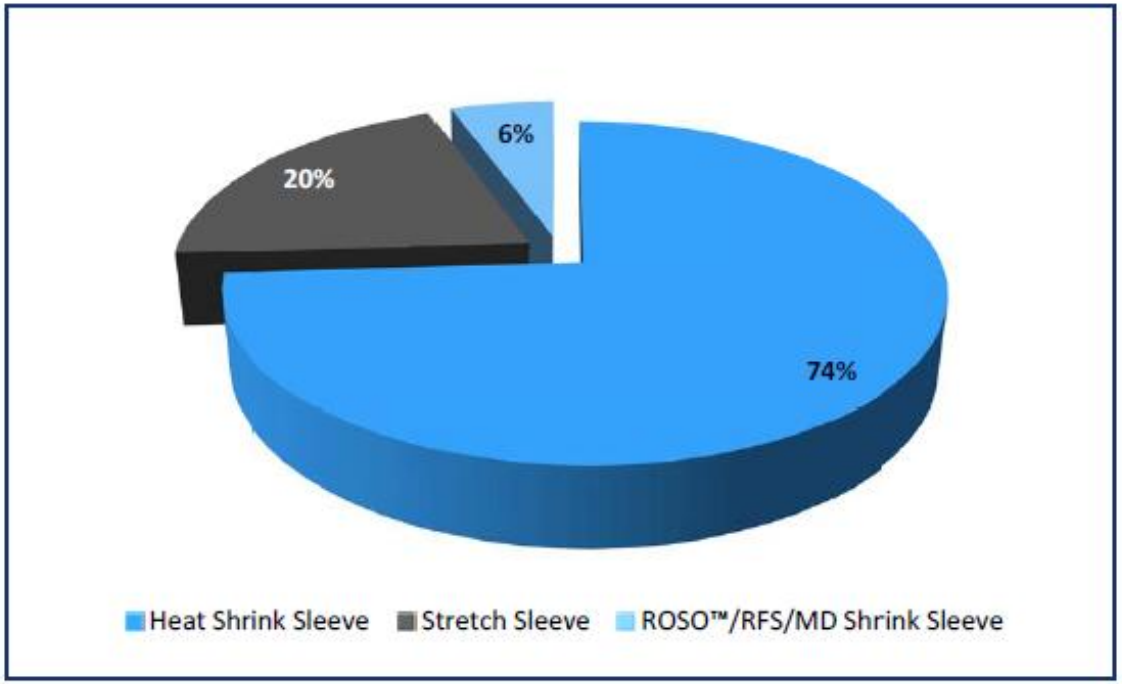
North America

	Low Shrink			Medium Shrink		
	Film Name	**Thickness** (µ)	**Yield** (m2/kg)	**Film Name**	**Thickness** (µ)	**Yield** (m2/kg)
Clear	510 CLS	15	72.5	440 CMS	18	62.6
	415 CLS	19	59	255 CMS	30.5	36.3
	306 CLS	25	43.5			
	255 CLS	31	36.3			
White Voided	370 WLS	31	52.6	370 WMS	31	52.6
	325 WLS-B	34	46.2	270 WMS	42	38.4
Metalized	370 WLSM	31	52.6			
	325 WLSM	34	46.2			

Market Overview
Shrink Label films

Taghleef Industries

Europe



Source: AWA

- WAL Shrink = 6% share of contoured container labeling; Metal can labeling less than NA
- One (1) Suppliers of WAL Shrink: ExMo
- No white voided film available
- Key Market Applications:
  - PET Beverage
  - Drinkable Yoghurt

Our products and applications

Europe

Taghleef Industries

	IN MOULD LABEL	SELF ADHESIVE LABEL	WRAP AROUND LABEL - Reel Fed	WRAP AROUND LABEL - Cut&Stack	WRAP AROUND LABEL - Shrink
Transparent	LIT	LSA, LNS, LSR	LTS, LTN, LTU	LCT	
Matt			LUS		
Solid White	LIW	LSW			
White voided	LIM, LIG, LIH, LFE	LSE	LRD, LGL, LXI	LCS	U002
Metallised		LZP	LZR, LZL		

NEW to EU

U002

- Developed based on the commercialized technology from North America
- White Voided Shrink Film now available to European Market

- Two (2) Films for Evaluation
  - 45µm **U002**
  - 53µm **U002**

Products and applications

Taghleef Industries

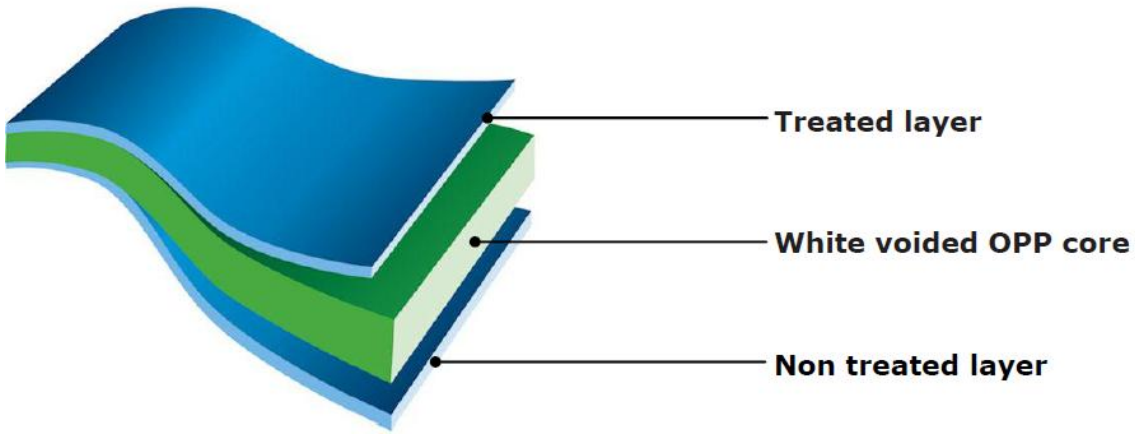
Region	Container	Top	Middle	Bottom	Rib	Max % Shrink	
Europe	San Benedetto (0.5L)	58.5	65.3	63.5	61.7	10.4%	
	Powerade Ion4 (500ml)	59.0	65.7	59.2	63.1	10.2%	
	Gatorade (0.5L)	56.7	63.0	58.4	60.5	10.0%	
	Danone Actimel	39.0	44.0	41.0	n/a	11.4%	
US	Powerade (20oz)	65.7	71.8	63.5	69.5	11.6%	
	Powerade (32oz)	84.5	90.0	78.5	85.7	12.8%	
	Lipton Tea Mix	91	103.2	91	n/a	11.8%	

Unit=mm

Data Sheet
European Shrink Label films :U002

Taghleef Industries

U002
(DRAFT VERSION)



Properties	Method	Unit	Ref.	Typical values	
Thickness	Internal method	µm		45	53
Unit weight		g/m²		27.5	31.9
Yield		m²/kg		36.4	31.3
Tensile strength	ASTM D882	N/mm²	MD TD	70 90	
Elongation at break		%	MD TD	70 45	
Dynamic cof	ASTM D1894		NT/NT	0.60	0.55
Thermal shrinkage	OPMA TC4(a)	%	MD	16	

Benefits of White Voided Film
Label films

Taghleef Industries

- Higher Yield Film
- Eliminate up to two (2) applications of white ink
- Eliminate “water-cling” look

- More vibrant graphics
- No hot-melt look

Future Developments

Taghleef Industries

- Expansion of our Label business:

  - ✓High shrink films (over 50%)
  - ✓**High yield** films with superior mechanical properties